CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

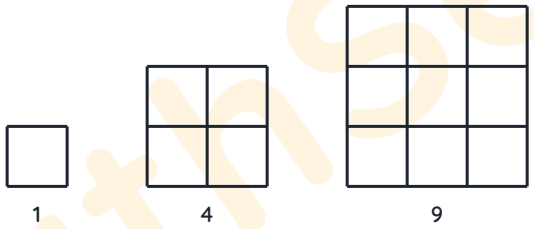
Chapter 1: A Square and A Cube

Topic-wise short notes with solved examples

Square numbers and cube numbers, the patterns they follow, and how to find square roots and cube roots using prime factorisation and estimation.

1 Square numbers

- A number multiplied by itself gives a **square number**: $n \times n = n^2$, read "n squared".
- The squares of natural numbers, $1, 4, 9, 16, 25, 36, \ldots$, are called **perfect squares**.
- The name comes from geometry: a square of side n units has area n^2 square units.
- Fractions and decimals can be squared too: $\left(\frac{3}{5}\right)^2 = \frac{9}{25}$ and $(2.5)^2 = 6.25$.



Squares of side 1, 2 and 3 contain 1, 4 and 9 unit squares

n	11	12	13	14	15	16	17	18	19	20
n^2	121	144	169	196	225	256	289	324	361	400

2 Factors and squares

- Factors come in pairs: $12 = 1 \times 12 = 2 \times 6 = 3 \times 4$, so 12 has 6 factors.
- In a perfect square one pair has two equal numbers ($36 = 6 \times 6$), so that factor is counted only once.
- **Only perfect squares have an odd number of factors.** Every other number has an even number of factors.

Example 2.1 Lockers numbered 1 to 50 are all closed. Person 1 changes every locker, person 2 changes every 2nd locker, person 3 every 3rd locker, and so on up to person 50. Which lockers are open at the end?

SOLUTION

Locker k is changed once for every factor of k.

It ends up open only if it is changed an odd number of times, that is, only if k has an odd number of factors.

Lockers 1, 4, 9, 16, 25, 36 and 49

3 Patterns in the units digit

- A perfect square ends in **0, 1, 4, 5, 6 or 9**. It never ends in 2, 3, 7 or 8.
- The test works one way only: 26 ends in 6 but is not a square. The units digit can show that a number is **not** a square; it cannot prove that it is.

Units digit of the number	1 or 9	2 or 8	3 or 7	4 or 6	5	0
Units digit of its square	1	4	9	6	5	0

- If a number ends in k zeros, its square ends in $2k$ zeros. So a perfect square ends in an **even** number of zeros.
- The square of an even number is even, and the square of an odd number is odd.

Example 3.1 Without calculating, say which of 4372, 9000, 5329 and 6258 cannot be perfect squares.

SOLUTION

4372 ends in 2 and 6258 ends in 8, so they are not squares.

9000 ends in three zeros, an odd number of zeros, so it is not a square.

4372, 9000 and 6258 are not perfect squares. (5329 may be one; in fact $5329 = 73^2$.)

4 Squares and odd numbers

- Differences of consecutive squares are the odd numbers: $4 - 1 = 3$, $9 - 4 = 5$, $16 - 9 = 7$.
- **The sum of the first n odd numbers is n^2**: $1 + 3 + 5 + 7 = 16 = 4^2$.
- The nth odd number is $2n - 1$.
- So $(n + 1)^2 = n^2 + (2n + 1)$: to go from one square to the next, add the next odd number.
- **Test by subtraction**: subtract $1, 3, 5, \ldots$ in turn. If you reach exactly 0 after k steps, the number is k^2. If you jump past 0, it is not a perfect square.
- Between n^2 and $(n + 1)^2$ there are exactly $2n$ numbers.

Example 4.1 Given $45^2 = 2025$, find 46^2.

SOLUTION

46^2 is the sum of the first 46 odd numbers, so add the 46th odd number to 45^2.

The 46th odd number is $2 \times 46 - 1 = 91$.

$46^2 = 2025 + 91 = 2116$

Example 4.2 How many numbers lie between 24^2 and 25^2?

SOLUTION

There are $2n$ numbers between n^2 and $(n+1)^2$. Here $n = 24$.

48

- **Triangular numbers** are $1, 3, 6, 10, 15, \ldots$ The sum of two consecutive triangular numbers is a square: $3 + 6 = 9$, $6 + 10 = 16$, $10 + 15 = 25$.

5 Square roots

- If $y = x^2$, then x is a **square root** of y. Finding a square root undoes squaring.
- Since $8^2 = 64$ and $(-8)^2 = 64$, the number 64 has two square roots, $+8$ and -8.
- The symbol $\sqrt{\ }$ stands for the **positive** square root: $\sqrt{64} = 8$. In this chapter only the positive root is used.
- **Prime factorisation method**: write the number as a product of primes. If the primes can be put into **pairs** with none left over, it is a perfect square. Take one prime from each pair and multiply.

Example 5.1 Find $\sqrt{1764}$.

SOLUTION

$1764 = 2 \times 2 \times 3 \times 3 \times 7 \times 7$

Every prime is paired, so $1764 = (2 \times 3 \times 7)^2$.

$\sqrt{1764} = 42$

Example 5.2 Is 1008 a perfect square? If not, find the smallest number it must be multiplied by to make it one.

SOLUTION

$1008 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 7$

The 7 has no partner, so 1008 is not a perfect square.

Multiply by 7. Then $7056 = (2 \times 2 \times 3 \times 7)^2 = 84^2$.

Example 5.3 Find the smallest square number that is divisible by 6, 8 and 15.

SOLUTION

The LCM of 6, 8 and 15 is $120 = 2^3 \times 3 \times 5$.

To complete the pairs, multiply by $2 \times 3 \times 5 = 30$.

3600

6 Estimating square roots

- Find the two perfect squares the number lies between. Then use the units digit and a midpoint to narrow the search.
- For a number that is not a perfect square, the square root lies between two consecutive whole numbers.

Example 6.1 Find $\sqrt{3364}$ by estimation.

SOLUTION

$50^2 = 2500$ and $60^2 = 3600$, so the root is between 50 and 60.

The number ends in 4, so the root ends in 2 or 8: it is 52 or 58.

$55^2 = 3025 < 3364$, so the root is more than 55.

$\sqrt{3364} = 58$ **(check: $58^2 = 3364$)**

Example 6.2 Between which two whole numbers does $\sqrt{200}$ lie? Which is it closer to?

SOLUTION

$14^2 = 196$ and $15^2 = 225$

Between 14 and 15, and much closer to 14

Example 6.3 A square sheet has an area of 500 sq cm. What is the largest square with a whole-number side (in cm) that can be cut from it?

SOLUTION

$22^2 = 484$ and $23^2 = 529$. Since $484 < 500 < 529$, the side of the sheet is between 22 cm and 23 cm.

A square of side 22 cm

7 Cube numbers

- A number multiplied by itself three times gives a **cube**: $n \times n \times n = n^3$, read "$n$ cubed".
- $1, 8, 27, 64, 125, \ldots$ are **perfect cubes**. A cube of edge n units is made of n^3 unit cubes.

n	1	2	3	4	5	6	7	8	9	10
n^3	1	8	27	64	125	216	343	512	729	1000

- A cube can end in **any** digit. The units digit of n^3 depends only on the units digit of n:

Units digit of n	0	1	2	3	4	5	6	7	8	9
Units digit of n^3	0	1	8	7	4	5	6	3	2	9

- The cube of a negative number is negative: $(-6)^3 = -216$. Also $\left(\frac{2}{3}\right)^3 = \frac{8}{27}$.
- If a number ends in k zeros, its cube ends in $3k$ zeros. So a cube cannot end in exactly two zeros.
- **Cubes and odd numbers**: $1 = 1^3$, $3 + 5 = 2^3$, $7 + 9 + 11 = 3^3$, $13 + 15 + 17 + 19 = 4^3$. The next n odd numbers always add up to n^3.

REMEMBER

1729 is the Hardy–Ramanujan number: the smallest number that is a sum of two cubes in two different ways, $1729 = 1^3 + 12^3 = 9^3 + 10^3$. Such numbers are called taxicab numbers.

8 Cube roots

- If $y = x^3$, then x is the **cube root** of y, written $\sqrt[3]{y}$. For example $\sqrt[3]{8} = 2$ and $\sqrt[3]{1000} = 10$.
- **Prime factorisation method**: if the prime factors can be put into **triplets** with none left over, the number is a perfect cube. Take one prime from each triplet and multiply.

Example 8.1 Find $\sqrt[3]{5832}$.

SOLUTION

$5832 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3$

Triplets: $(2 \times 2 \times 2) \times (3 \times 3 \times 3) \times (3 \times 3 \times 3)$

$\sqrt[3]{5832} = 2 \times 3 \times 3 = 18$

Example 8.2 What is the smallest number by which 675 must be multiplied to get a perfect cube?

SOLUTION

$675 = 3 \times 3 \times 3 \times 5 \times 5$

The 3s form a triplet. The two 5s need one more 5.

5 (and $3375 = 15^3$)

Example 8.3 13824 is a perfect cube. Guess its cube root without factorising.

SOLUTION

It ends in 4, so the cube root ends in 4.

Drop the last three digits: 13. Since $2^3 = 8 < 13 < 27 = 3^3$, the tens digit is 2.

24

9 Common mistakes

- Deciding that a number is a perfect square because it ends in 0, 1, 4, 5, 6 or 9. The units digit only rules numbers out.
- Thinking n^2 means $2 \times n$, or n^3 means $3 \times n$.
- Leaving an unpaired prime inside a square root, or forgetting that cube roots need triplets and not pairs.
- Writing $\sqrt{64} = \pm 8$. The symbol means the positive root only.
- Saying a cube cannot end in 2, 3, 7 or 8. That rule is for squares.

MathSetu